Choosing to type this answer as my hands are starting to not work (accommodations include this I think?); my apologies for the change in format!

1. What is the “technology independent” phase of logic synthesis?

The “technology independent phase of logic synthesis describes the point in design development that is not dependent on the technology chosen, IE where the code will be written or what chip the code will be run on. It is quite early on in the logic synthesis process, as it is done pre-synthesis and pre-anything concrete!

2. What is the cost metric used for a circuit?

The cost metric used to value a circuit is the “sum of the literals in the factored form of the node expressions”.

“The cost given is that of the resulting network, as measured by the literal count of the nodes expressed in factored form.” - The Synthesis of Cyclic Combinational Circuits paper
“For each node, we expect the lowest cost expression to be obtained with the full substitutional set (i.e., all other node functions) and the highest cost expression to be obtained with the empty set” -The Synthesis of Cyclic Combinational Circuits paper

I can’t find anywhere else to mention how entertained I am by the concept of the Jiggle Algorithm mentioned in a cited paper.
R. K. Brayton, G. D. Hachtel, C. T. McMullen, and A.
L. Sangiovanni-Vincentelli, “Multilevel Logic
Synthesis,” Proceedings of the IEEE, Vol. 78, No. 2,
p. 264 – 300, 1990.

3. Describe substitution.

“Node functions are expressed, or re-expressed, in terms of other node functions as well as of their original inputs.”-The Synthesis of Cyclic Combinational Circuits paper

4. Why would you want to get rid of cycles in combinational circuits?

“The behavior of a circuit with feedback is generally more complicated. Such a circuit may exhibit timing-dependent behavior (as in the case of an R-S Latch), and it may be unstable (as in the case of an oscillator).”-The Synthesis of Cyclic Combinational Circuits paper

If there is no specific benefit for the circuit to be cyclic, then it is better for the circuit to be simpler, “loop-free”, and more “clearly combinational”. -The Synthesis of Cyclic Combinational Circuits paper

5. What is topological vs. electric feedback?

Topological feedback refers to a visible loop in logic that is nevertheless ironed out when real logic values are introduced; see the ADECA cycle in Figure two, but how it is smoothed out by adding values in Figure 4.

Electrical feedback in a cyclic circuit refers to how outputs may not settle on an electrical value due to feedback from a loop, called the “Up-Bounded inertial delay model” citing J. A. Brzozowski and C.-J. H. Seger, “Asynchronous Circuits,” Springer-Verlag, 1995.

(Figures from The Synthesis of Cyclic Combinational Circuits paper)